

UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF M.Sci. WITH HONOURS

FOURTH-YEAR FINAL EXAMINATION

03 00563

LM GENERAL THEORY OF RELATIVITY

SEMESTER 1 EXAMINATIONS 2022/23

Time allowed: 1 hour 30 minutes

*Answer **two** questions. If you answer more than two questions, credit will only be given for the best two answers.*

The approximate allocation of marks to each part
of a question is shown in brackets [].

All symbols have their usual meanings.

A formula sheet will be found at the end of this question paper.

No calculator is permitted in this examination.

Answer **two** questions. If you answer more than two questions, credit will only be given for the best two answers.

1. Consider the motion of a particle of unit mass in the *equatorial plane* around a black hole.

(a) Using the Schwarzschild metric given in the attached formulae sheet, write down the corresponding Lagrangian *for the planar motion* and formulate three conservation laws, including those for the energy, $\mathcal{E}$, and the angular momentum, J , of the particle. [6]

(b) Use the conservation laws to prove that the proper radial velocity of the particle is given in terms of $\mathcal{E}$ and J as follows:

$$\dot{r}^2 = \frac{\mathcal{E}^2}{c^2} - V_{\text{eff}}(r),$$

where $V_{\text{eff}}(r)$ is the effective potential which you need to find. [3]

(c) Show that the above equation can be written in terms of the dimensionless variables, $\varepsilon \equiv \mathcal{E}/c^2$, $j \equiv J/(r_s c)$, and $\chi \equiv r_s/r$ with $0 < \chi < 1$ as

$$\frac{\dot{r}^2}{c^2} = \varepsilon^2 - \tilde{V}(\chi), \quad \tilde{V}(\chi) = (1 - \chi)(1 + j^2 \chi^2),$$

where $\tilde{V}(\chi)$ is the effective dimensionless potential. What is the minimal energy of a particle at a radial coordinate $r = r_s/\chi$? [3]

Now consider various types of motion which are possible for a particle with the dimensionless angular momentum $j = \sqrt{27/8}$.

(d) Find the radiuses of stable and unstable circular orbits and corresponding dimensionless energy ε of the particle in these orbits. [9]

[Hint: It might be easier to find min and max of $\tilde{V}(\chi)$ for an arbitrary j and substitute the given value of j into this.]

(e) Find at what dimensionless energies

- all the particles with $\dot{r} > 0$ escape while all with $\dot{r} < 0$ fall into the black hole;
- all the particles fall into the black hole, independently of the sign of $\dot{r}$;
- closed particle orbits exist at some values of χ which you do not need to specify.

Illustrate your answers graphically, sketching $\tilde{V}(\chi)$ on the $\varepsilon - \chi$ plane. [9]

2. Consider a two-dimensional curved space.

(a) Argue that the Riemann tensor $R^\mu_{\nu\alpha\beta}$, (with $\mu, \nu = 1, 2$) may have no more than one independent nonzero component. [4]

(b) Although two-dimensional surfaces provide convenient means for modeling motion in a curved space, they cannot be used for constructing a direct analogue of Einstein's equations in empty space. Give arguments to support this statement. [4]

(c) Show that in coordinates where the metric tensor, $g_{\mu\nu}$, is diagonal, the scalar curvature, R , may be expressed in terms of $R^\mu_{\nu\alpha\beta}$ as

$$R = 2g^{22}R^1_{212} = 2g^{11}R^2_{121}. \quad [4]$$

Consider a two-dimensional curved surface with the metric

$$ds^2 = \frac{dr^2}{r(1-r)} + 4(1-r) d\phi^2,$$

where the contravariant coordinates $r \equiv x^1$ and $\phi \equiv x^2$ are defined on the intervals $0 < r < 1$ and $0 \leq \phi < 2\pi$.

(d) Write down nonzero components of $g_{\mu\nu}$ and $g^{\mu\nu}$. [2]

(e) Show by direct calculation that all nonzero Christoffel symbols are given by

$$\Gamma^1_{11} = \frac{1}{2} \left(\frac{1}{1-r} - \frac{1}{r} \right), \quad \Gamma^1_{22} = 2(r - r^2), \quad \Gamma^2_{21} = \Gamma^2_{12} = -\frac{1}{2(1-r)}. \quad [9]$$

(f) By using the above expression for R and expressing $R^\mu_{\nu\alpha\beta}$ in terms of the Christoffel symbols, or otherwise, find the scalar curvature R . [7]

3. The Friedman - Robertson - Walker metric of a homogeneous and isotropic universe is

$$ds^2 = c^2 dt^2 - \mathcal{R}^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\theta^2 + r^2 \sin^2\theta d\phi^2 \right]$$

Here r is a dimensionless radial coordinate, $\mathcal{R}(t)$ is a history function, and k is a curvature parameter: $k = \pm 1$ for positively or negatively curved spaces, or $k=0$ for a flat space.

Consider a model of an **expanding** universe where the radiation contribution is neglected and $\mathcal{R}(t)$ and k are to be found from the Einstein–Friedman equations:

$$\frac{\dot{\mathcal{R}}^2 + kc^2}{\mathcal{R}^2} = \frac{8\pi G\rho_M(t)}{3} + \frac{\Lambda c^2}{3}, \quad \frac{\ddot{\mathcal{R}}}{\mathcal{R}} = -\frac{4\pi G\rho_M(t)}{3} + \frac{\Lambda c^2}{3}.$$

Here $\rho_M(t)$ is the matter density and the time-independent parameter Λ represents the vacuum (dark energy) contribution with the energy density $\rho_V = \Lambda c^2/4\pi G$.

- (a) Rewrite these equations by expressing $\dot{\mathcal{R}}/\mathcal{R}$ and $\ddot{\mathcal{R}}/\mathcal{R}$ in terms of the Hubble ‘constant’ H and ‘deceleration’ parameter q . [5]
- (b) Formulate, in terms of the Hubble ‘constant’ $H(t_V) \equiv H_V$ in a certain epoch t_V , the conditions satisfied by $\rho_M(t_V)$ and Λ for the universe to be flat in this epoch. [5]
- [Hint: You might find using the density parameter Ω helpful.]
- (c) Next, formulate the condition satisfied by $\rho_M(t_V)$ for the universe to be vacuum-dominated and remain flat. [3]
- (d) Neglecting the matter contribution, find the ‘deceleration’ parameter q and show that $H = H_V$ for $t \geq t_V$. Explain why H remaining constant is consistent with nonzero acceleration of the universe expansion. [8]
- (e) Calculate the history function, $\mathcal{R}(t)$, and the matter density, $\rho_M(t)$, for $t \geq t_V$, assuming the initial conditions $\rho_M(t = t_V) = \rho_0$ and $\mathcal{R}(t = t_V) = \mathcal{R}_0$. [9]

Formulae Sheet

Unless otherwise stated, all indices take values of $\mu = (0, m) \equiv (0, 1, 2, 3)$ in the GR context, or $\mu = (1, \dots, d)$ in a generic d -dimensional space, and the Einstein summation convention is implied

Metric and Lagrangian

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu, \quad \mathcal{L} \equiv g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu, \quad \dot{x}^\mu \equiv u^\mu \equiv \frac{dx^\mu}{ds}, \quad g_{\mu\nu} \text{ is a metric tensor}$$

Here s is any affine parameter along a geodesic line. In GR context, $s \mapsto \tau$ where τ is the proper time

The Schwarzschild metric

$$ds^2 = c^2 \sigma(r) dt^2 - \sigma^{-1}(r) dr^2 - r^2 [d\theta^2 + \sin^2\theta d\phi^2], \quad \sigma(r) \equiv \left(1 - \frac{r_s}{r}\right), \quad r_s \equiv \frac{2GM}{c^2}$$

Christoffel symbols, $\Gamma_{\alpha\beta}^\mu$

$$\Gamma_{\alpha\beta}^\mu \equiv g^{\mu\nu} \Gamma_{\nu\alpha\beta}, \quad \Gamma_{\nu\alpha\beta} = \frac{1}{2} (g_{\nu\alpha,\beta} + g_{\nu\beta,\alpha} - g_{\alpha\beta,\nu})$$

Covariant derivatives

$$\nabla_\beta \xi^\mu \equiv \xi^\mu_{;\beta} = \xi^\mu_{,\beta} + \Gamma_{\alpha\beta}^\mu \xi^\alpha, \quad \nabla_\beta \xi_\mu \equiv \xi_{\mu;\beta} = \xi_{\mu,\beta} - \Gamma_{\mu\beta}^\alpha \xi_\alpha, \quad \xi^\mu_{,\beta} \equiv \partial_\beta \xi^\mu \equiv \frac{\partial \xi^\mu}{\partial x^\beta}$$

Covariant dot operation

$$\dot{\xi} \equiv \frac{d}{ds} (\xi^\alpha \mathbf{e}_\alpha) = (\nabla_u \xi^\alpha) \mathbf{e}_\alpha \equiv (\xi^\alpha_{,\mu} + \Gamma_{\nu\mu}^\alpha u^\nu) u^\mu \mathbf{e}_\alpha \text{ where } \nabla_u \equiv u^\mu \nabla_\mu$$

Geodesic equations in the Euler–Lagrange form

$$\frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\alpha} \right) - \frac{\partial \mathcal{L}}{\partial x^\alpha} = 0$$

Geodesic equations in canonical form

$$f^\alpha = \frac{g^{\alpha\beta}}{2} \left[\frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\beta} \right) - \frac{\partial \mathcal{L}}{\partial x^\beta} \right] = \ddot{x}^\alpha + \Gamma_{\mu\nu}^\alpha \dot{x}^\mu \dot{x}^\nu = 0 \Leftrightarrow \dot{u}^\alpha + \Gamma_{\mu\nu}^\alpha u^\mu u^\nu = 0 \Leftrightarrow \nabla_u u^\alpha = \nabla_u^2 x^\alpha = 0.$$

The Riemann, $R_{\nu\alpha\beta}^\mu$

$$R_{\nu\alpha\beta}^\mu \xi^\nu = \xi^\mu_{;\alpha\beta} - \xi^\mu_{;\beta\alpha}, \quad R_{\nu\alpha\beta}^\mu = -\Gamma_{\nu\alpha,\beta}^\mu + \Gamma_{\nu\beta,\alpha}^\mu - \Gamma_{\nu\alpha}^\sigma \Gamma_{\beta\sigma}^\mu + \Gamma_{\nu\beta}^\sigma \Gamma_{\alpha\sigma}^\mu.$$

The Riemann in locally (pseudo)Cartesian frame (LCF)

$$g_{\mu\nu} R^{\mu}_{\alpha\beta\gamma} = R_{\nu\alpha\beta\gamma} = g_{\nu\gamma,\alpha\beta} - g_{\nu\beta,\alpha\gamma} + g_{\alpha\beta,\nu\gamma} - g_{\alpha\gamma,\nu\beta}.$$

Symmetries of the Riemann tensor

$$R^{\alpha}_{\beta\mu\nu} + R^{\alpha}_{\mu\nu\beta} + R^{\alpha}_{\nu\beta\mu} = 0, \quad R^{\alpha}_{\beta\mu\nu} = -R^{\alpha}_{\beta\nu\mu}, \quad R_{\alpha\beta\mu\nu} = R_{\mu\nu\alpha\beta}.$$

Geodesic deviation

$$f^{\alpha} = \nabla_u^2 x^{\alpha}(\tau; \eta_0 + d\eta) = R^{\alpha}_{\beta\mu\nu} u^{\beta} u^{\mu} \xi^{\nu}, \quad u^{\mu} = \frac{dx^{\mu}(\tau; \eta_0)}{d\tau}, \quad \xi^{\nu} = \frac{dx^{\nu}(\tau; \eta)}{d\eta} d\eta.$$

The Ricci, $R_{\mu\nu}$, and Einstein, $G_{\mu\nu}$, tensors and scalar curvature, R

$$R_{\mu\nu} \equiv R^{\alpha}_{\mu\alpha\nu}, \quad G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R, \quad R = g^{\mu\nu} R_{\mu\nu}.$$

Einstein's equations

$$G_{\mu\nu} = \kappa T_{\mu\nu} \iff R_{\mu\nu} = \kappa \left(T_{\mu\nu} - \frac{1}{2} g_{\mu\nu} T^{\alpha}_{\alpha} \right); \quad \kappa \equiv \frac{8\pi G}{c^4}, \quad T_{\mu\nu} \text{ is a matter tensor.}$$

In the cosmological context, $\Lambda g_{\mu\nu}$ should be added to the r.h.s., where $\Lambda > 0$ is the cosmological constant.

The matter tensor in the perfect fluid model of the Universe

$$T^{\mu}_{\nu} = (\rho_M + \rho_R) u^{\mu} u_{\nu} - P \delta^{\mu}_{\nu}.$$

Here $\rho_{M(R)}$ is the matter (radiation) density, with $\rho_R \equiv 3P/c^2$ where P is the radiation pressure.

The Friedmann–Robertson–Walker metric

$$ds^2 = c^2 dt^2 - \mathcal{R}^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2\theta d\phi^2) \right].$$

Here $k = 1, 0$, or -1 is the curvature parameter, and $\mathcal{R}(t)$ is the history function.

The Hubble ‘constant’ H and ‘deceleration’ parameter q

$$H \equiv \frac{\dot{\mathcal{R}}}{\mathcal{R}}, \quad q \equiv -\frac{\ddot{\mathcal{R}}}{\dot{\mathcal{R}}^2}, \quad \text{where} \quad \dot{\mathcal{R}} \equiv \frac{d\mathcal{R}}{dt}, \quad \ddot{\mathcal{R}} \equiv \frac{d^2\mathcal{R}}{dt^2}.$$

Density parameter

$$\Omega = \Omega_M + \Omega_R + \Omega_V = \frac{\rho_M + \rho_R + \rho_V}{\rho_{\text{crit}}}, \quad \text{where} \quad \rho_V \equiv \frac{\Lambda c^2}{8\pi G} \quad \text{and} \quad \rho_{\text{crit}} \equiv \frac{3H^2}{8\pi G}.$$

NO CALCULATOR

This page is intentionally blank.

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.